Surveillance Poisons Communication in LLM Global Games

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Abstract

In global games of regime change, citizens receive noisy private signals about regime strength and must independently decide whether to join an uprising, knowing it succeeds only if enough others join too. Theory predicts a unique threshold equilibrium, but testing how information structures shape coordination has proven difficult: laboratory experiments strip away the richness of real information environments, and field data confound strategic behavior with institutional differences. I embed seven large language models in the Morris–Shin (2003) regime-change game, conveying private signals as natural-language intelligence briefings rather than numeric values. Across 1,800 country–periods, LLM agents display robust threshold behavior aligned with the theoretical benchmark (mean $r = +0.80$, $p < 0.001$ for every model). The central finding concerns surveillance. Adding a monitoring warning to the pre-play communication stage—while leaving the decision prompt unchanged—lowers joining by -9.7 pp across five models. The mechanism is communication poisoning: agents self-censor their messages, degrading informativeness (R^2 drops from 0.12 to 0.02), and a degraded-messages control without surveillance framing produces an even larger suppression. The same channel that helps citizens coordinate is also the channel an authoritarian regime can exploit.

JEL: C72, C92, D82, D83, P16
Keywords: global games, regime change, surveillance, communication, LLM agents, preference falsification

1 Introduction

The information channel is a trap. In the canonical global game of regime change (Morris and Shin, 2003), private signals alone resolve equilibrium multiplicity (Carlsson and van Damme, 1993; Frankel et al., 2003)—but real coordination crises, from bank runs (Diamond and Dybvig, 1983) to currency attacks (Obstfeld, 1996) to political upheaval (Angeletos et al., 2007), also feature communication among agents. Any channel that transmits information about others' willingness to act is also the channel through which regimes suppress it. This paper augments the Morris–Shin framework with an explicit pre-play communication network and documents how surveillance exploits that channel.

The idea behind a global game is simple. A regime has some underlying strength, but no citizen knows it exactly. Each citizen receives a private, noisy signal—some see evidence of weakness, others see evidence of stability—and must decide whether to join an uprising or stay home. Joining is risky: it pays off only if enough others join too. Staying home is safe but forgoes the chance of change. The strategic tension is that each citizen's best action depends on what they believe *others* will do, which in turn depends on what others believe, and so on. Carlsson and van Damme (1993) and Morris and Shin (2003) showed that when citizens receive sufficiently precise private signals, this infinite regress collapses to a unique equilibrium: there is a single threshold above which citizens stay and below which they join. This result transformed the study of coordination failures—currency crises, bank runs, political revolutions—by providing a tractable framework with sharp comparative statics.

Testing these predictions empirically is difficult. Laboratory experiments confirm threshold behavior with stylized numeric signals, but cannot manipulate the rich, multi-source information that characterizes real crises. Field data from actual coordination failures confounds strategic behavior with institutional differences and unobserved information structures. The gap between the theory's tractability and the difficulty of testing it empirically motivates this paper's approach.

I address this gap by embedding seven large language models (LLMs) spanning six architecture families in the Morris and Shin (2003) regime change game. Each agent receives a private signal $x_i = \theta + \varepsilon_i$, translated into a natural-language intelligence briefing whose content shifts monotonically with signal strength across eight evidence domains. No explicit payoff table is provided—the stakes are embedded in the narrative, forcing agents to form beliefs from language.

Contribution roadmap. The paper's core contribution is the surveillance result. I show that the communication channel that enables coordination is also exploitable by an authoritarian planner. Surveillance degrades message informativeness (R^2 drops from 0.12 to 0.02) and lowers joining by -9.7 pp across five models. A degraded-messages control—replacing LLM-generated messages with generic uninformative text, without any surveillance framing—drops joining by 21 pp, confirming that the main mechanism operates through communication poisoning rather than direct deterrence. Elicited beliefs shift significantly under surveillance, with a residual belief–action wedge consistent with preference falsification.

The surveillance claim rests on a behavioral foundation. Across 1,800 country–periods and seven models, LLM agents display robust threshold-like behavior aligned with the global-game benchmark (mean $r = +0.80$, $p < 0.001$ for every model); a narrative stakes ladder confirms that LLMs respond to strategic framing, not just text sentiment (Appendix E.10).

Related literature. The paper connects two strands. The global game literature—from Carlsson and van Damme (1993) through Morris and Shin (2003) and the dynamic extensions of Angeletos et al. (2007) and Chassang (2010)—provides the theoretical framework; laboratory experiments (Heinemann et al., 2004, 2009; Szkup and Trevino, 2020) confirmed threshold behavior with numeric signals. The political economy of authoritarian control—Edmond (2013) on propaganda, Kuran (1991) on preference falsification, Shadmehr and Bernhardt (2011) on collective action—motivates the surveillance experiments. Empirical evidence that censorship targets coordination (King et al., 2013), surveillance suppresses expression (Penney, 2016; Stoycheff, 2016), and social media enables protest (Enikolopov et al., 2020) provides the real-world stakes.[1]

On the methodological side, Horton (2023) proposed LLMs as "homo silicus." Subsequent work found LLMs struggle in coordination games (Akata et al., 2025; Sun et al., 2025), with model scale not predicting strategic performance (Petrov et al., 2025). Huang et al. (2024) and Carlini et al. (2025) document that alignment shifts risk preferences, motivating narrative rather than tabular payoffs. Critical reviews by Larooij and Törnberg (2025a,b) identify validation challenges that I address through cross-generator robustness (Appendix E.7), scramble/flip falsification, and out-of-sample replication.

The paper proceeds as follows. Section 2 presents the theoretical framework. Section 3 describes the experimental design. Section 4 establishes a threshold-like behavioral benchmark and shows that pre-play communication creates an informative channel. Section 5 tests whether an authoritarian regime can exploit that channel through surveillance. Section 6 concludes.

[1] Concurrent work by Ruano and Rajan (2026) places LLM depositors in a bank-run coordination game with network contagion, finding sharp phase transitions consistent with Diamond–Dybvig.

2 The Global Game of Regime Change

A continuum of citizens indexed by $i \in [0,1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta \leq 0$ represent regimes so weak they fall without opposition; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ (Eq. 1):

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i \, di > \theta. \tag{1}$$

Payoffs depend on the citizen's action and the outcome (Eq. 2):

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases} \tag{2}$$

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Nonparticipants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003)**.** *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium in threshold strategies. An agent joins if and only if $x_i < x^*$ (Eq. 3), where*

$$x^* = \theta^* + \sigma \Phi^{-1}(\theta^*) \tag{3}$$

and $\theta^* = B/(B+C)$.

The *attack mass* $A(\theta)$—the theoretical fraction of the population that joins at regime strength θ—is given by:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right). \tag{4}$$

This is a decreasing function of θ: weaker regimes face larger uprisings.

I define $J(\theta)$ as the *empirical join fraction*—the finite-sample, observed counterpart to the theoretical attack mass $A(\theta)$. My experiment uses $n = 25$ agents with a normal prior $\theta \sim \mathcal{N}(\bar{z}, 1)$, and agents do not observe (B, C, σ) directly. I use the continuum $A(\theta)$ as an external behavioral benchmark throughout, not as the exact equilibrium of the implemented finite-n game.

The briefing generator—the system that converts z-scores into natural-language intelligence briefings—has three tunable control parameters (clarity, directional precision, dissent framing). Full generator details are in Appendix E.11.

3 Experimental Design

The experiment has two phases. First, I establish that LLM agents display threshold-like policies in the global game when private signals are conveyed in natural language: a pure treatment (private signals only), a communication treatment (pre-play messaging), and falsification tests. Second, I test whether an authoritarian regime can exploit the communication channel through surveillance. All LLM interactions use the same prompt structure across models.

In the theory (Section 2), θ is an unbounded fundamental. In the benchmark experiments, θ is drawn from a normal distribution and agents are not shown payoff parameters (B, C). I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$ ($\theta^* = 0.50$, $\sigma = 0.3$).

Throughout, $\theta^* = B/(B + C)$ is the theoretical cutoff, $x^* = \theta^* + \sigma\Phi^{-1}(\theta^*)$ the signal cutoff, and $\hat{\theta}^*$ the estimated cutoff from the fitted logistic $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$, where $b_1 > 0$ corresponds to a join curve that is *decreasing* in θ (the expected direction). The cutoff is $\hat{\theta}^* = -b_0/b_1$. I write $\beta \equiv b_1$ for the logistic steepness parameter in tables.

For each country–period, nature draws $\theta \sim \mathcal{N}(\bar{z}, 1)$, where $\bar{z}$ is a public prior mean drawn randomly for each country. Each agent i receives a private signal $x_i = \theta + \varepsilon_i$. The platform normalizes that signal to $z_i = (x_i - \bar{z})/\sigma$ and translates z_i into a multi-paragraph intelligence briefing via a deterministic generator that maps signal strength to narrative content about regime stability, economic conditions, public sentiment, and coordination prospects. Agents observe the briefing, not the normalization inputs $(\bar{z}, \sigma)$ themselves. Figure 1 summarizes the signal-to-text-to-decision pipeline.

Agents observe only their private briefing and never the payoff parameters (B, C), the noise level σ, or the prior distribution. The diffuse-prior equilibrium (Proposition 1) serves as an external benchmark for behavioral shape and comparative statics, not as a claim about the exact equilibrium of the implemented game.

The behavioral benchmark has four core treatments: a *pure global game* (independent decisions), *communication* (pre-play messaging on a Watts–Strogatz network, $k = 4$, $p = 0.3$), *scramble* (cross-period briefing permutation breaking the θ-to-signal link), and *flip* (negated z-scores). A *degraded-messages* control replaces LLM messages with generic text, providing an exogenous benchmark for communication quality. Additional robustness tests are in Appendix E.

I test seven models spanning six families (Table 1). All experiments use $n = 25$ agents per country–period and $\sigma = 0.3$. Benchmark experiments use $B = C = 1$ ($\theta^* = 0.50$). All LLM calls use temperature $= 0.7$ with a single sample per decision, routed through OpenRouter and cached by request hash (Appendix F). Throughout, n denotes agents per country–period and N denotes the number of country–periods.

Table 1: Model summary. Columns report country-period counts in the pure, communication, and falsification (scramble+flip) suites. All runs use $N = 25$ agents per period and $\sigma = 0.3$.

Model	Arch.	Pure	Comm	Falsif.
Mistral Small Creative	Mistral	1000	1000	300
Llama 3.3 70B	Llama	100	100	200
Qwen3 30B	Qwen (MoE)	100	100	200
GPT-OSS 120B	GPT	200	200	1000
Qwen3 235B	Qwen (MoE)	200	200	200
Trinity Large	Arcee	100	100	200
MiniMax M2-Her	MiniMax	100	100	200
Total		**1800**	**1800**	**2300**

Notes: 7 models spanning 6 architecture families. N = country-periods with 25 agents each. All runs use $\sigma = 0.3$ and temperature $= 0.7$ unless otherwise stated.

The unit of randomization is the country–period (θ draw plus agent-level decoding stochasticity). For agent-level regressions, standard errors are clustered at the model–country–period level. For period-level correlations, I report Fisher-z confidence intervals.

The surveillance treatment augments the communication prompt with a monitoring warning; the decision prompt is unchanged (full prompts in Appendix B).

4 Behavioral Benchmark

Result 1 (Threshold-Policy Alignment)**.** *Across seven models and 1,800 country–periods in the pure global game treatment, the Pearson correlation between the empirical join fraction and the theoretical attack mass $A(\theta)$ (Eq. 4) averages $r = +0.80$ ($p < 0.001$ for every model).*[2]

Figure 2 shows the empirical sigmoid for the primary model. Correlations span $r \in [+0.75, +0.84]$ across architectures (Table 2); the pooled OLS is $J = 0.13 + 0.54\,A(\theta)$ ($R^2 = 0.48$). Model-specific action biases shift the intercept but not the correlation (Figure A9).

Result 2 (Signal Dependence)**.** *Cross-period scrambling of briefings reduces the mean within-country correlation from $r = +0.80$ to $r = +0.03$ across seven models. The pooled correlation drops from $r = +0.76$ to $r = +0.01$ (Fisher $z = 26.17$, $p < 0.001$).*

Result 3 (Signal Direction)**.** *Inverting the signal direction flips the mean correlation from $r = +0.80$ to $r = -0.80$ across seven models. The pooled correlation moves from $r = +0.76$ to $r = -0.79$ (Fisher $z = 53.85$, $p < 0.001$).*

The pure → scramble → flip pattern replicates across all seven models (Figure 3; Table 2).

[2] The primary model (Mistral) was run in multiple batches with `--append`, producing 1,000 rows from 680 unique (country, period, θ) configurations. Averaging within each configuration (deduplication) changes r from 0.753 to 0.739; all qualitative results are unaffected.

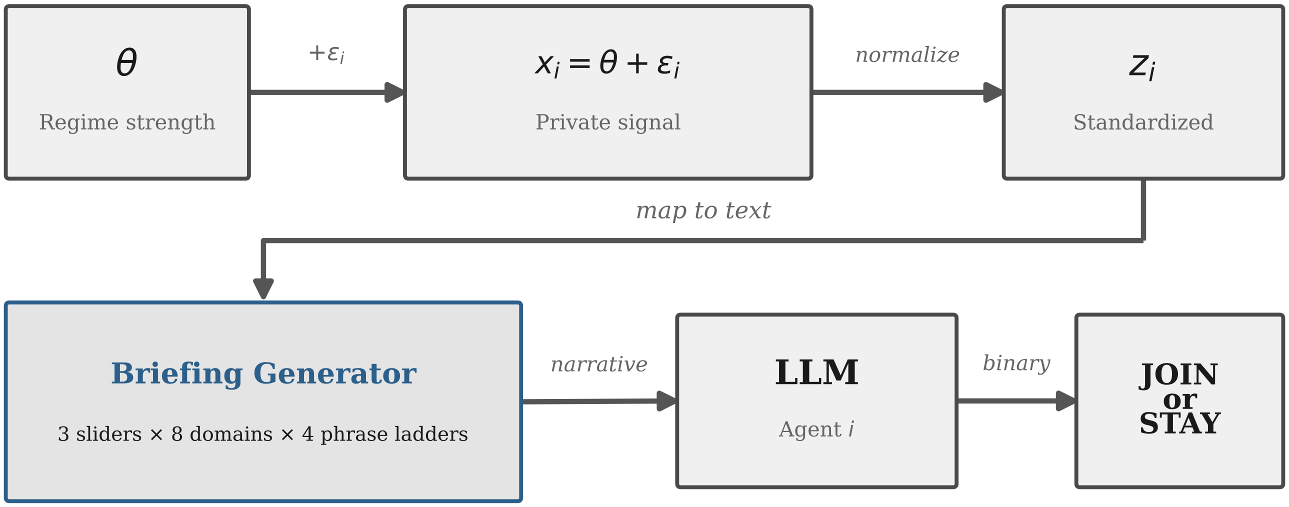

Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i, converted to z-scores and rendered into natural-language briefings via 8 evidence domains and 3 latent sliders. The briefing layer is deterministic; all stochasticity enters through LLM decoding.

Table 2: Threshold-policy alignment by model and treatment. Cells report Pearson r between the empirical join fraction and the benchmark attack mass $A(\theta)$ under $B = C = 1$ (so $\theta^* = 0.50$); 95% Fisher-z confidence intervals in brackets for main treatments.

	Main treatments		Falsification			
Model	Pure	Comm	Scramble	Flip	n_{pure}	Mean join
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	−0.03	−0.81	1000	0.387
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02	−0.84	100	0.439
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13	−0.87	100	0.499
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	−0.01	−0.80	200	0.407
Qwen3 235B	+0.80 [0.74, 0.84]	+0.76 [0.70, 0.81]	+0.06	−0.85	200	0.416
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05	−0.80	100	0.464
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01	−0.62	100	0.436
Pooled	+0.76 [0.74, 0.78]	+0.75 [0.73, 0.77]	+0.01	−0.79	1800	0.409
Mean across models	+0.80	+0.78	+0.03	−0.80	—	—

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass under $B = C = 1$. 95% Fisher-z confidence intervals in brackets. Join fraction uses valid decisions only (excluding parse errors). Pooled: all country-periods concatenated; Mean: equal-weighted average across models.

A text-only baseline tracks LLM decisions at $r = 0.79$, but the LLM's join curve is substantially steeper (Figure A10). An eight-condition *narrative stakes ladder*—prepending headers from "existential consequences" to "zero personal cost" to identical briefings—shifts mean joining monotonically from 21% to 79% (57.6 pp), with paraphrase variants tracking their originals and a text classifier unable to detect the header (Appendix E.10).

Communication.

Result 4 (Communication Creates an Informative Channel)**.** *Matched-cell comparison across $n = 1479$ unique (model, country, period, θ) task cells yields $\Delta = +2.10$ pp ($t = 6.383$, $p < 0.001$). Communication modestly increases joining in like-for-like tasks.*

Communication preserves the signal structure ($r = +0.78$ vs. $+0.80$ under pure; Figure 4). Regressing θ on message features yields $R^2 = 0.12$—messages carry real information, noisily. This informativeness is what makes the channel exploitable: surveillance degrades R^2 to 0.02 (Section 5), an 81% reduction.

A *degraded-messages* control replaces all LLM-generated messages with generic uninformative text while preserving the two-stage protocol and omitting any surveillance

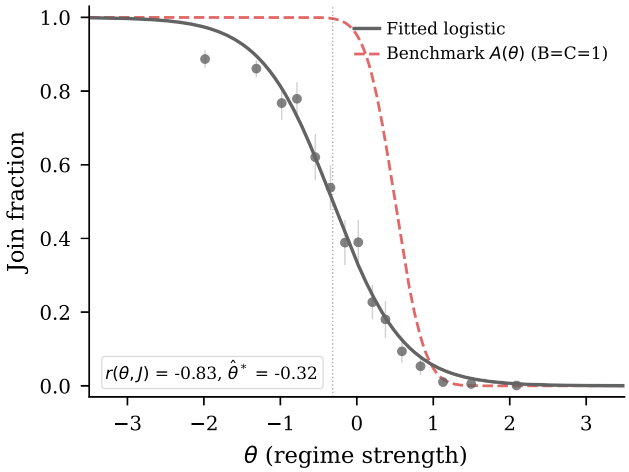

Figure 2: Empirical join fraction vs. θ (Mistral, 1,000 periods). The empirical sigmoid is shifted leftward ($\hat{\theta}^* = -0.32$) relative to theory ($\theta^* = 0.50$). Cross-model results in Table 2 (mean $r = +0.80$, all $p < 0.001$).

framing. Against its own live-messages baseline ($n = 500$, mean join 45.9%), degraded messages drop the mean join rate to 24.5% ($\Delta = -21.4$ pp, $p < 0.001$)—well below even the pure game (38.7%)—confirming that the communication benefit is driven by informative content, and that any mechanism degrading message quality should produce a comparable suppression.

5 Surveillance

In the surveillance treatment, the communication prompt is augmented with a warning that communications are being monitored by regime security services. The manipulation affects only the communication phase; the decision prompt is unchanged. The isolation is architectural: each LLM call is stateless—the decision-stage API call receives only the briefing and the (now self-censored) messages, with no memory of the communication-stage surveillance warning. Any difference in join rates must therefore operate through the *content* of the messages that the decision stage receives.

Result 5 (Surveillance Produces a Large Chilling Effect)**.** *Surveillance reduces mean join rates in the primary model (Mistral Small Creative) from 41.1% to 27.6%, a shift of* -13.5 *pp (*$p < 0.001$*). The action–attack-mass correlation is preserved (*$r = +0.68$ *vs.* $r = +0.74$ *under the communication baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.*

Surveillance contaminates the information environment with self-censored messages, pushing join rates well below the communication baseline (Figure 5). The effect replicates across five models: Mistral (-13.5 pp), GPT-OSS (-11.9 pp), Qwen3 30B (-10.9 pp), Llama (-8.9 pp), and Qwen3 235B (-3.3 pp), averaging -9.7 pp. Two placebo variants—monitored for research (no consequences) and aggregated anonymously—produce no significant deviation from the communication baseline ($+2.6$ pp and $+4.0$ pp, both $p > 0.2$; Appendix E.4).

The mechanism is communication poisoning. Three pieces of evidence identify communication degradation as the primary channel.

First, the R^2 of message content on true θ drops from 0.12 to 0.02—an 81% reduction in informational content. Message content confirms self-censorship: "act" drops from 15.8% to 4.1% of messages; "collapse" drops from 9.6% to 0.6%. Among agents who privately decide to JOIN, 61.0% send action-signaling messages under surveillance vs. 80.0% under regular communication.

Second, the degraded-messages control (Section 4, Communication) exogenously replaces all messages with generic text—without any surveillance framing—and drops joining by 21.4 pp (to 24.5% from its 45.9% live-messages baseline), a *larger* suppression than the 13.5 pp drop under surveillance. This is the expected pattern if surveillance operates primarily by degrading message informativeness: the exogenous control removes all information, while self-censorship under surveillance only partially degrades it.

Third, elicited punishment risk is $\approx 8.0/10$ across all conditions with negligible differences between JOIN and STAY agents (Table A16), supporting the contaminated-information interpretation over a direct deterrence channel.

Beliefs and preference falsification. Post-decision beliefs, elicited on the information set agents held at decision time (including peer messages; $n = 12{,}500$ per condition), drop -5.7 pts under surveillance ($p < 0.001$); second-order beliefs drop -8.2 pts ($p < 0.001$). Actions shift further (-11.4 pp)—a residual wedge consistent with Kuran's (1991) preference-falsification logic (full factorial in Appendix C, Table A2).

A separate hold-fixed test replays baseline messages verbatim while adding a surveillance warning to the *decision* prompt, producing a -20.9 pp drop ($p < 0.001$). This confirms a direct decision-stage channel exists—but it is absent in the main design, where the decision prompt contains no surveillance reference. The main treatment isolates the indirect channel: communication poisoning.

6 Conclusion

The paper's central result is that surveillance poisons the communication channel. When monitoring enters only during the messaging stage, join rates fall sharply, message informativeness drops from 0.12 to 0.02, placebo warnings do not replicate the effect, and an exogenous degraded-messages control produces an even larger suppression. The

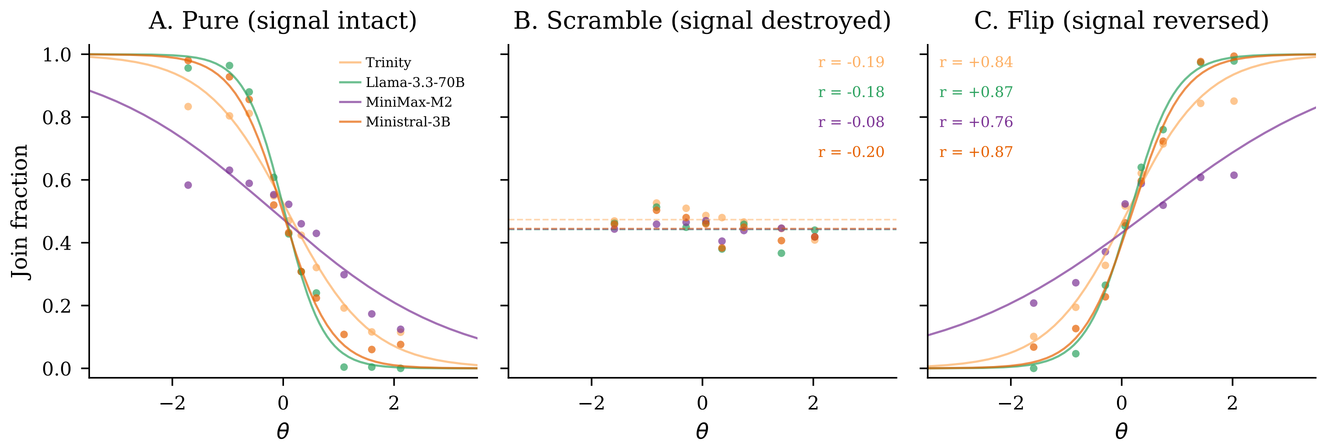

Figure 3: Falsification triptych. *Left:* Pure (mean $r = +0.80$). *Center:* Scramble ($r = +0.03$). *Right:* Flip ($r = -0.80$).

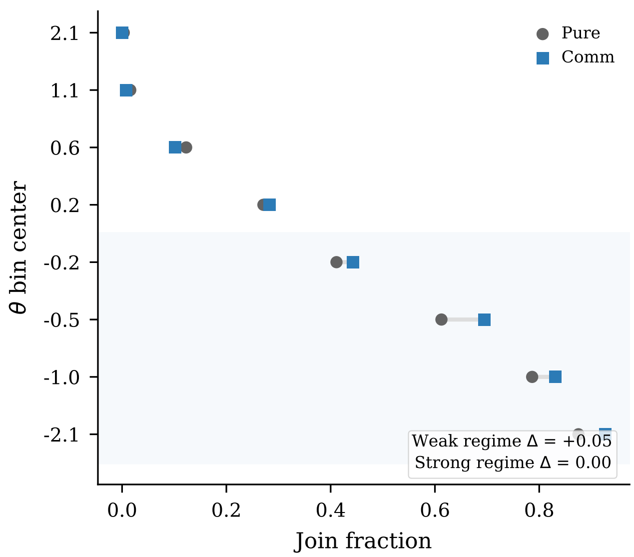

Figure 4: Communication effect by regime strength, pooled across seven models.

same channel that helps citizens coordinate is also the channel an authoritarian regime can degrade.

The paper avoids claiming full recovery of the unique Bayesian Nash equilibrium or causal mediation through post-decision belief elicitations. Three limitations warrant discussion. First, whether the underlying mechanism is approximate Bayesian reasoning or a learned heuristic that matches BNE remains open. Second, the experiments are static; real coordination problems are dynamic (Angeletos et al., 2007; Chassang, 2010). Third, external validity remains the fundamental challenge: what LLM agents do in a coordination game is evidence about the structure of the game, not necessarily about how humans would play it. The cross-model replication (seven architectures, five surveillance variants) suggests that the qualitative patterns are properties of the information structure rather than artifacts of particular model training. The experimental template is reusable, and the data and code are publicly available.

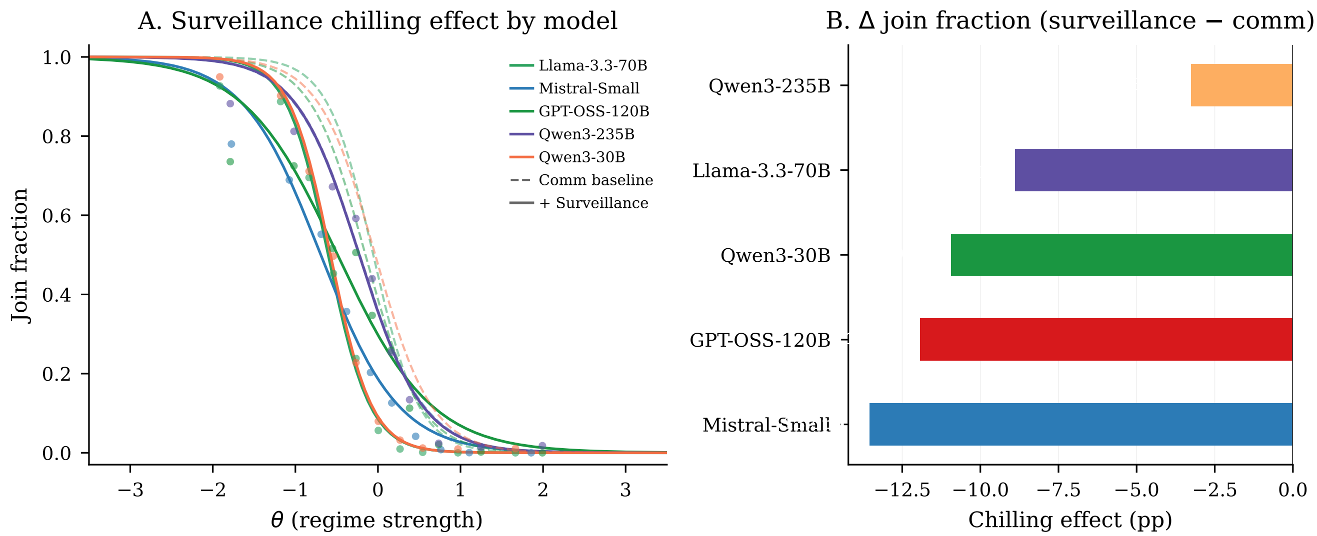

Figure 5: Surveillance reduces join rates across five models, averaging −9.7 pp.

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A Experimental Design Details

Table A1 reports eight pre-specified hypotheses and their outcomes. The main-text results correspond to H1 (alignment), H3 (flip), H4 (communication), and H7 (surveillance); all four survive Bonferroni correction at $\alpha = 0.00625$. H5–H6 and H8 test information design and propaganda treatments reported in supplementary materials.

Narrative header texts. The eight narrative stakes-ladder conditions prepend the following headers to each agent's standard intelligence briefing (the briefing body is identical across conditions):

> **Very high cost:** "CONTEXT: The regime maintains an extensive network of informants and has a documented history of disappearances following failed uprisings. Participants' families are routinely targeted. The cost of unsuccessful action is existential."
>
> **Moderate high cost:** "CONTEXT: Participants in past failed uprisings have faced prison sentences of several years and loss of employment. The personal cost of unsuccessful action is significant but not life-threatening."
>
> **Neutral:** "CONTEXT: The consequences of participating in a failed uprising are uncertain and vary case by case. Some participants have faced penalties while others have not. The personal stakes are ambiguous."
>
> **Moderate low cost:** "CONTEXT: Recent legal reforms have limited the regime's ability to punish dissent. Past participants faced only brief questioning and were released without charges. The personal risk of action is modest."
>
> **Very low cost:** "CONTEXT: The international community has guaranteed safe passage and asylum for all participants regardless of outcome. Even failed uprisings carry essentially zero personal cost."
>
> **Placebo:** "CONTEXT: The country has experienced significant weather disruptions this year, including severe flooding in the eastern provinces and an extended drought in agricultural regions. Infrastructure damage has been substantial."

Paraphrase variants (high cost v2, low cost v2) use different surface wording to express the same cost level.

B Complete Prompt Example

The following shows the complete text that an LLM receives for a single decision. The *decision prompt* is:

> "You are a citizen deciding whether to JOIN an uprising or STAY home. You only have your private briefing below. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning."

This is followed by the agent's intelligence briefing (see Figure A12 for complete examples at $z \in \{-2, 0, +2\}$). In the *communication* treatment, agents first receive:

> "Based on the intelligence briefing you received, write a brief message (1–2 sentences) to share your read of the situation with your trusted contacts."

After exchanging messages on the network, agents receive the decision prompt above, augmented with: "Before deciding, you received the following messages from your contacts:" followed by the actual messages from network neighbors. In the *surveillance* variant, the communication prompt appends:

> "You have reason to believe that your communications are being monitored by regime security services. Messages deemed subversive could have serious consequences."

The decision prompt is *unchanged* under surveillance—it contains no mention of monitoring. Any behavioral effect therefore operates through the content of the (now self-censored) messages.

C Belief Elicitation

Beliefs are elicited on a 0–100 probability scale immediately after each decision ("How likely is it that the regime falls this period?"). The primary elicitation conditions on the full information set agents held at decision time: the private briefing, the agent's choice, and the peer messages received in the communication round. Figure A1 documents calibration quality against the theoretical posterior ($r = +0.78$); Figure A3 confirms monotone signal–belief mapping nonparametrically.

A restricted variant excluding peer messages tests information-set sensitivity. Table A2 reports the 2×2 factorial crossing condition (communication vs. surveillance) with information set (messages included vs. excluded). Without messages, beliefs barely shift under surveillance (-0.4 pts, $p = 0.34$) while actions shift -11.4 pp. With messages included, beliefs shift significantly (-5.7 pts, $p < 0.001$): roughly half the apparent belief–action wedge under the restricted design was a measurement artifact. Figure A2 shows second-order beliefs under the restricted information set.

D Identification Test Details

Table A3 compares text-classifier accuracy under baseline and surveillance conditions, establishing that the chilling

Table A1: Pre-specified hypotheses and test results. H1–H4 use pooled Part I data across all seven models; H5–H8 use the primary model (Mistral Small Creative). Effect size: r for correlations (H1–H3), Cohen's d or d_z for mean comparisons (H4–H8). "Supported" at $\alpha = 0.05$.

H	Hypothesis	Test	Stat	p	Effect	Supported?
H1	Sigmoid Response	Pearson	0.757	<0.001	0.757	Yes
H2	Scramble Falsification	Pearson	0.014	0.638	0.014	Yes
H3	Directional Sensitivity	Pearson (1-sided)	-0.791	<0.001	-0.791	Yes
H4	Communication Channel	Paired t	6.383	<0.001	0.166	Yes
H5	Ambiguity Pooling	t-test	-0.828	0.408	-0.071	No (ambiguous)
H6	Censorship Distortion	t-test	-12.207	<0.001	-1.051	Yes
H7	Surveillance Chilling	t-test	-8.277	<0.001	-0.370	Yes
H8	Propaganda Dose-Response	t-test	-0.892	0.372	-0.049	No (ambiguous)

Notes: H1–H4: pooled across all seven models (H4 uses paired t-test matched on model/country/period/θ task cells). H5–H8: primary model (Mistral Small Creative). Bonferroni survivors at $\alpha = 0.00625$: H1, H3, H4, H6, H7. H8 post-hoc power: 0.14 (Cohen's $d = -0.049$; MDE at 80% power = 6.3 pp).

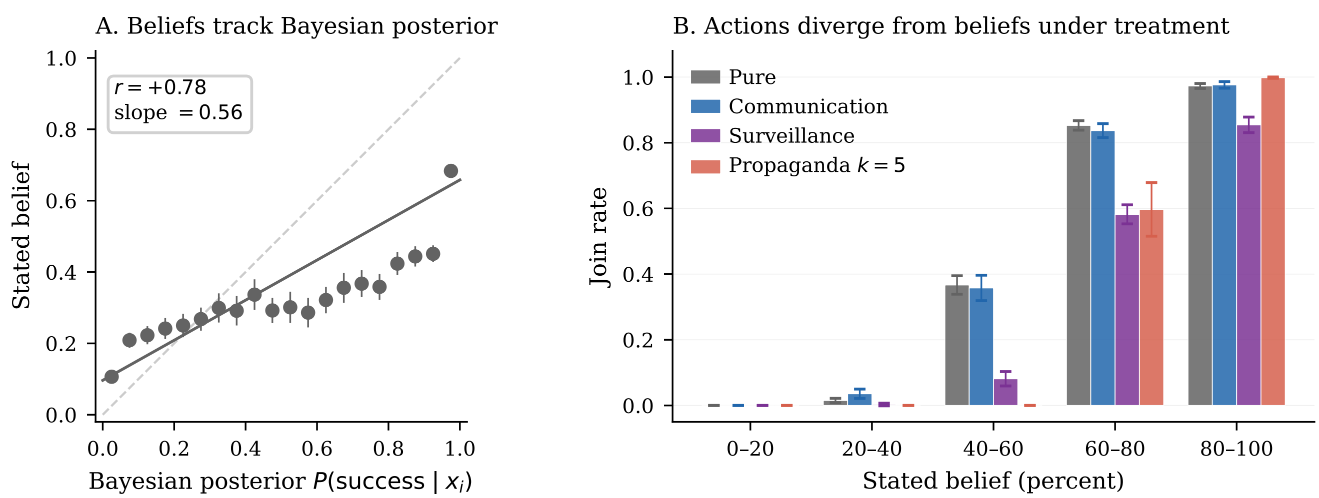


Figure A1: Belief elicitation (Mistral). *Left:* Stated beliefs vs. theoretical $P(\text{success} \mid x_i)$ ($r = +0.78$). *Right:* Join rate by belief bin.

Table A2: Belief–action decomposition under surveillance: effect of including peer messages in the belief elicitation prompt (n = 12,500 per cell).

	Messages excluded		Messages included	
	Comm	Surv	Comm	Surv
First-order belief	46.0	45.7	53.2	47.4
Second-order belief	32.7	32.6	42.0	33.8
Join rate	42.4%	31.0%	42.2%	30.8%
Surveillance effect (comm → surv):				
Δ belief	−0.4 ($p = 0.34$)		−5.7 ($p < 0.001$)	
Δ SOB	−0.0 ($p = 0.89$)		−8.2 ($p < 0.001$)	
Δ join	−11.4 pp ($p < 0.001$)		−11.4 pp ($p < 0.001$)	

Notes: Post-decision elicitation, 500 country–periods × 25 agents per cell. "Messages excluded": belief prompt sees briefing and decision only. "Messages included": belief prompt additionally sees the peer messages received in the communication round.

Table A3: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. "Pred. join (surv.)" is the classifier's predicted join rate when applied to surveillance-treatment briefings; "Actual" is the LLM's observed rate. The gap measures the surveillance wedge invisible to text classifiers.

Classifier	Acc.	AUC	Pred. join (surv.)	Actual (surv.)	Gap
BoW TF-IDF	88.4%	0.947	40.5%	27.6%	13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	-19.2 pp

Notes: Primary model (Mistral Small Creative). Accuracy and AUC from 5-fold cross-validation on pure-treatment data. Gap = classifier-predicted join rate − actual LLM join rate under surveillance (pp).

effect is invisible to classifiers restricted to the intelligence briefing body alone. Table A4 repeats the classifier exercise across the cost/benefit sweep.

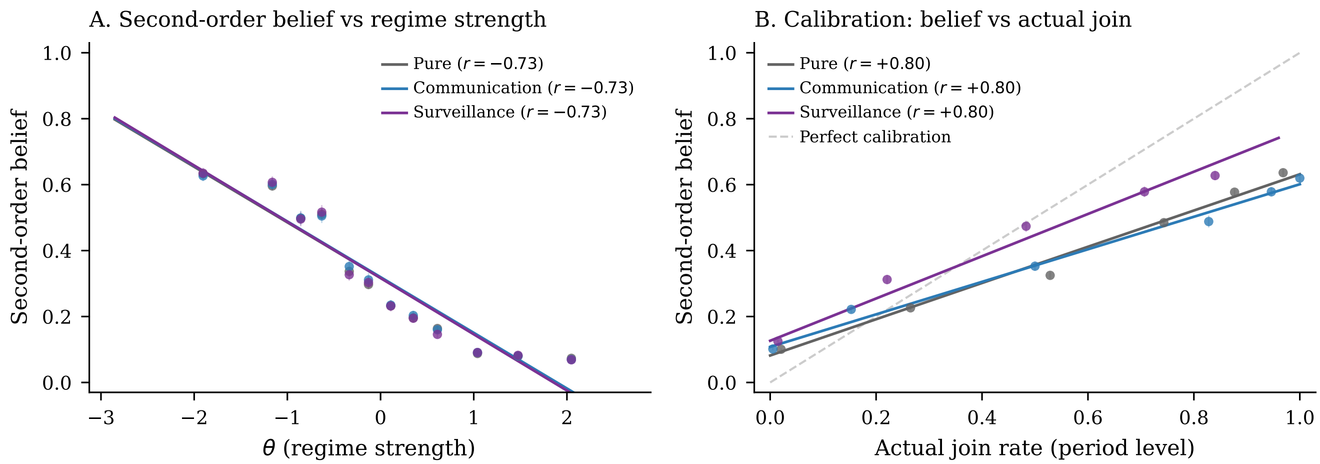


Figure A2: Second-order beliefs (restricted information set, messages excluded). On this information set, surveillance and pure overlap; the factorial in Table A2 shows a significant shift when messages are included.

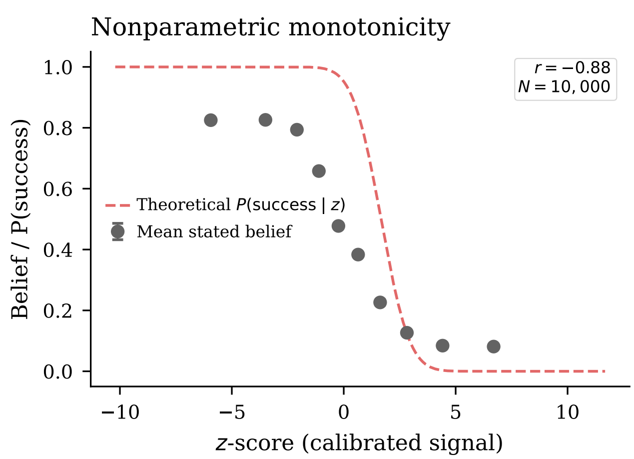
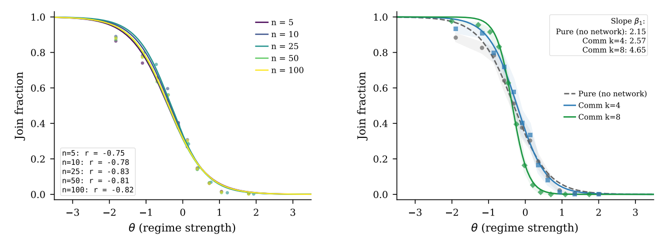


Figure A3: Nonparametric monotonicity: mean belief by z-score bin vs. theoretical benchmark.



(a) Agent count.

(b) Network density.

Figure A4: Robustness checks for threshold-policy alignment.

Table A4: Narrative-stakes conditions: classifier vs. actual LLM behavior. Actual join rates shift by ≈ 50 pp across strategic-stakes framing conditions; the classifier, trained on briefing-body text features alone, cannot predict this shift.

Condition	N	Classif. pred.	Actual	Gap (pp)
Baseline ($\theta^* = 0.50$)	270	40.9%	40.9%	+0.0
High cost ($\theta^* = 0.25$)	270	40.9%	19.0%	+21.9
Low cost ($\theta^* = 0.75$)	270	40.9%	69.3%	-28.4

Notes: Primary model (Mistral Small Creative). Logistic classifier trained on baseline slider features. Gap = classifier-predicted join rate − actual LLM join rate (pp).

E Robustness

Threshold-policy alignment is stable to agent count ($r \in [+0.60, +0.65]$ across $n \in \{5, 10, 50, 100\}$), network density ($r = +0.66$ at $k = 8$ vs. $+0.74$ at $k = 4$), and mixed-model populations ($r = +0.77$ pure, $r = +0.75$ communication; Figure A4).

E.1 Communication Aggregation

Communication aggregation accounting. Table A5 decomposes the communication estimators by model. The matched-cell estimand is the main one because it compares exact like-for-like task cells and loses almost no support; the gap relative to the pooled unpaired contrast is driven by weighting repeated task cells, especially in the largest model, rather than by a large common-support restriction.

Without calibration, all 6 models show $r > 0.71$ and 5 exceed $r > 0.75$ (Table A6). Deliberately miscalibrating two models (± 0.3 shift) leaves r virtually unchanged ($r \in [+0.75, +0.76]$; Table A7): calibration adjusts the *level* but does not create the *slope*. Table A8 and Figure A6 report substantial cross-model heterogeneity in calibration quality, with r_A ranging from the mid-0.6s to the mid-0.8s.

Table A5: Why the communication estimators differ.

Model	Rows/arm	Row wt. (P/C)	Cells/arm	Matched	Cell wt.	Lost (P/C)	Δ unpaired	Δ paired
Mistral Small Creative	1000	55.6%	680	680	46.0%	0/0	+2.38	+4.84
Llama 3.3 70B	100	5.6%	100	100	6.8%	0/0	-2.36	-2.36
Qwen3 30B	100	5.6%	100	100	6.8%	0/0	-4.64	-4.64
GPT-OSS 120B	200	11.1%	200	200	13.5%	0/0	+3.47	+3.47
Qwen3 235B	200	11.1%	200	200	13.5%	0/0	+0.12	+0.12
Trinity Large	100/99	5.6%/5.5%	100/99	99	6.7%	1/0	-3.14	-3.41
MiniMax M2-Her	100	5.6%	100	100	6.8%	0/0	+1.32	+1.32
Equal-weight avg	—	14.3% each	—	—	14.3% each	—	-0.41	-0.09
Pooled	1800/1799	100.0%	1480/1479	1479	100.0%	1/0	+1.23	+2.10

Notes: "Rows/arm" counts valid country-period rows entering the pooled unpaired estimator in each arm. "Cells/arm" counts unique task cells after collapsing duplicates by the exact matching key (model, country, period, θ, z, benefit, θ^*). The paired estimator averages within these cells and keeps only common support; "Row wt." and "Cell wt." are the model shares in the pooled unpaired and pooled paired estimators, respectively. Only Trinity Large loses support, with one pure-only cell and no communication-only cells, so the pooled gap is overwhelmingly a weighting/re-averaging issue rather than a support-loss issue.

Table A6: Uncalibrated robustness: models run without any calibration adjustment. Even without calibration, six of seven models show strong $r(J, A(\theta))$, confirming that the sigmoid is not an artifact of the calibration procedure.

Model	N	Mean join	$r(J, A(\theta))$	p
Mistral Small Creative	100	0.382	+0.81	0.0000
Llama 3.3 70B	100	0.422	+0.83	0.0000
Qwen3 30B	100	0.585	+0.90	0.0000
GPT-OSS 120B	100	0.336	+0.77	0.0000
Qwen3 235B	100	0.445	+0.86	0.0000
MiniMax M2-Her	100	0.425	+0.71	0.0000

Notes: Pure treatment, no calibration adjustment (cutoff center = 0). Trinity Large excluded due to elevated API error rates. $r = r(J, A(\theta))$.

Table A7: Placebo calibration. The cutoff center is deliberately shifted by ± 0.3 from its calibrated value. The action–attack-mass correlation $r(J, A(\theta))$ is unchanged; only the mean join rate shifts, confirming that calibration does not create the sigmoid.

Model	Condition	N	Mean join	$r(J, A(\theta))$
Mistral Small Creative	Calibrated	—	0.387	+0.75
	$\Delta c = +0.3$	100	0.424	+0.76
	$\Delta c = -0.3$	100	0.362	+0.76
Llama 3.3 70B	Calibrated	—	0.439	+0.82
	$\Delta c = +0.3$	100	0.385	+0.75
	$\Delta c = -0.3$	100	0.340	+0.76

Notes: Pure treatment. $\Delta c = \pm 0.3$: deliberate shift from calibrated cutoff center. Correlation $r(J, A(\theta))$ is unchanged; only the mean join rate shifts.

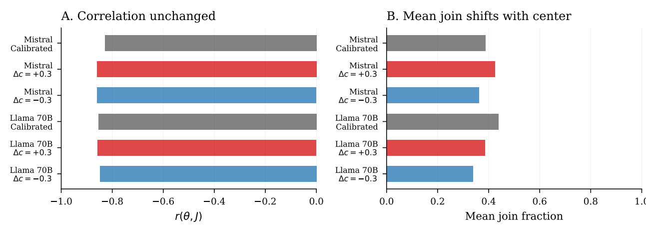


Figure A5: Placebo calibration: deliberately miscalibrating Mistral and Llama (± 0.3 shift) leaves r virtually unchanged, confirming calibration adjusts the *level* but does not create the *slope*.

Table A8: Calibration robustness. r_θ: raw correlation with regime strength. r_A: correlation with theoretical attack mass. RMSE: root mean squared error vs. $A(\theta)$. Text slope: logistic slope of naïve $1 -$ direction predictor.

Model	r_θ	r_A	RMSE	Text slope
Mistral Small Creative	−0.81	+0.67	0.354	-9.2
Llama 3.3 70B	−0.85	+0.79	0.288	-25.1
Qwen3 30B	−0.84	+0.78	0.287	-7.5
GPT-OSS 120B	−0.84	+0.70	0.359	-8.2
Qwen3 235B	−0.85	+0.70	0.354	-20.5
Trinity Large	−0.87	+0.84	0.262	-4.0
MiniMax M2-Her	−0.79	+0.66	0.360	-2.1

Notes: r_θ: Pearson correlation with regime strength θ. r_A: correlation with theoretical attack mass $A(\theta)$. Text slope: logistic slope of naïve $1 -$ direction predictor (the direction slider inverted as a baseline text classifier). All models use pure treatment with calibrated parameters.

E.2 Generator Robustness

Replacing the logistic function mapping z-scores to directional sentiment with three alternatives—tanh, piecewise linear, and a hard step function at $z = 0$—yields nearly identical correlations: $r = 0.77$ (tanh), $r = 0.77$ (linear), $r = 0.77$ (step) on $n = 100$ country-periods each, compared with $r = +0.75$ for the logistic baseline (main pure sample, $n = 1000$). A domain ablation test removes sub-

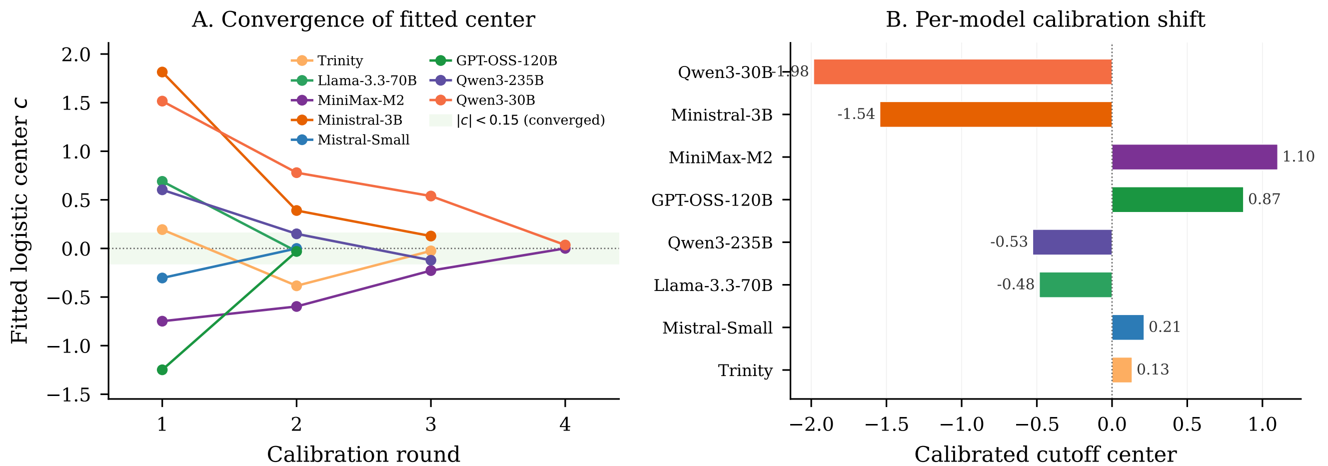


Figure A6: Calibration convergence. *Left:* Trajectory of fitted logistic center across autocalibration rounds. *Right:* Final calibrated cutoff center per model.

sets of the briefing generator's eight evidence domains. Removing the two coordination-language domains raises mean joining by 2.8 pp while preserving the threshold relationship ($r = 0.90$). Removing the four regime-state domains lowers mean joining by 3.4 pp and attenuates the threshold relationship ($r = 0.86$).

E.3 Level-k Cognitive Hierarchy Comparison

Human subjects in coordination games exhibit level-k cognitive hierarchy behavior rather than full equilibrium reasoning (Szkup and Trevino, 2020). I evaluate theoretical attack masses under BNE, L_1, and L_2 benchmarks. Across the seven payoff-sweep conditions, the fitted cutoff tracks the BNE benchmark more closely than L_1 or L_2 (RMSE $= 0.199$, $r = +0.74$ vs. $0.307/+0.45$ and $0.348/+0.35$), but the shifted-grid design means this evidence is best read as a centering diagnostic rather than a standalone rejection of level-k behavior.

The correlation $r(J, A(\theta))$ ranges from $+0.76$ to $+0.84$ across 13 model–temperature combinations ($T \in \{0.3, 0.5, 0.7, 1.0, 1.2\}$ for three models; Table A9).

E.4 Falsification and Placebo Tests

Two placebo variants—monitoring "for research purposes" (no consequences) and "aggregated anonymously"—produce no significant deviation from the communication baseline ($+2.6$ pp, $p = 0.4164$ and $+4.0$ pp, $p = 0.2006$; Table A10). The chilling effect is specific to the surveillance warning's strategic content.

E.5 Finite-n Aggregation Check

The continuum benchmark $\theta^* = B/(B + C)$ assumes a measure-zero agent and diffuse priors. With $n = 25$ discrete agents, the regime falls when $\sum a_i > n\theta$ (a binomial count), so the exact equilibrium object for the implemented game differs from the continuum limit. Table A11

Table A9: Temperature robustness across three models. The pure global game is run at varying LLM decoding temperatures. The correlation $r(J, A(\theta))$ is stable across all temperatures and models.

Model	T	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small	T=0.3	100	0.412	+0.78	-0.049
Mistral Small	T=0.7	100	0.406	+0.76	-0.079
Mistral Small	T=1.0	100	0.410	+0.79	-0.039
Llama 70B	T=0.3	100	0.364	+0.78	-0.210
Llama 70B	T=0.5	100	0.360	+0.78	-0.219
Llama 70B	T=0.7	100	0.361	+0.78	-0.220
Llama 70B	T=1.0	100	0.360	+0.78	-0.215
Llama 70B	T=1.2	100	0.358	+0.78	-0.227
Qwen 235B	T=0.3	100	0.392	+0.83	-0.117
Qwen 235B	T=0.5	100	0.392	+0.82	-0.112
Qwen 235B	T=0.7	100	0.392	+0.84	-0.110
Qwen 235B	T=1.0	100	0.398	+0.83	-0.105
Qwen 235B	T=1.2	100	0.397	+0.83	-0.121

Notes: Pure treatment, calibrated parameters, varying LLM decoding temperature across three models.

Table A10: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Variant	N	Mean join	$r(J, A(\theta))$	Δ (pp)	p
Placebo	200	0.437	+0.75	+2.6	0.416
Anonymous	200	0.452	+0.77	+4.0	0.201

Notes: Primary model (Mistral Small Creative). Placebo: agents told monitoring is for research purposes only (no consequences). Anonymous: messages aggregated before delivery. Δ: change vs. communication baseline (pp). p-value from two-sample t-test.

reports a narrower check: for each model, I fit a logistic to the observed join curve (70% train), then compute $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ as the predicted regime fall rate. Predicted rates correlate with empirical rates at $r \geq 0.95$ for every model (Mistral: $r = 0.9968$).

Table A11: Finite-N Benchmark: Predicted vs. Empirical Regime Fall Rates

Model	N periods	Logistic x_0	Pearson r	RMSE	MAE
Mistral	599	−0.11	0.997***	0.042	0.015
Llama 70B	99	0.05	0.954***	0.156	0.059
Qwen 30B	99	0.20	0.986***	0.088	0.029
GPT-OSS 120B	199	−0.01	0.997***	0.038	0.014
Qwen 235B	199	−0.03	0.989***	0.073	0.025
Trinity	99	0.19	0.979***	0.108	0.043
MiniMax	99	1.02	0.999***	0.028	0.013
Pooled	1399	−0.05	0.999***	0.019	0.007

Notes: For each θ bin, the predicted fall rate is $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability. Pearson r measures correlation between predicted and empirical fall rates across θ bins. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E.6 Agent-Level Analysis

Table A12 reports agent-level logit regressions (N = 289,555, clustered SEs). All treatment effects are significant and correctly signed. Marginal effects at the mean: a one-unit increase in θ reduces $P(\text{join})$ by 41 pp; surveillance reduces it by 36 pp. Column 3 estimates the association between elicited belief and action, controlling for the agent's z-score. **Caveat:** beliefs are elicited *post-decision*, so the near-perfect pseudo-R^2 (0.975) likely reflects post-decision rationalization rather than a causal pathway from beliefs to action.

Figure A7(A) shows a three-feature model (direction, clarity, coordination) outperforms a one-feature sentiment baseline; Panel (B) confirms cross-treatment generalization from pure to communication and surveillance.

E.7 Cross-Generator Robustness

Three text rendering formats—baseline narrative, diplomatic cable, journalistic wire—use identical slider functions but differ in surface prose. Correlations are virtually identical (max within-model difference 0.02; Figure A8, Table A14), ruling out prose style as the driver.

E.8 Belief Calibration and Risk Elicitation

Stated beliefs track the theoretical success probability ($r_{\text{belief,posterior}} = +0.78$) and predict actions ($r_{\text{belief,decision}} = +0.84$; partial $r = +0.64$ controlling for signal; Table A15). The surveillance belief shift and preference-falsification wedge are reported in the main text; the full factorial is in Appendix C.

Mean punishment risk is $\approx 8.0/10$ across all conditions with negligible JOIN/STAY differences (< 0.2 points; Table A16), supporting the contaminated-communication interpretation.

E.9 Cross-Model and Text Baseline Details

E.10 Stakes and Pipeline Centering

Table A17 reports the full conditions for the payoff sweep and narrative stakes ladder. Figure A11 plots the fitted cutoff $\hat{\theta}^*$ against the theoretical θ^* across the seven payoff configurations.

E.11 Briefing Generator Examples

Table A19 reports the three slider values (direction, clarity, coordination) at representative z-scores. The direction slider is logistic in z (slope 0.8, centered at 0). Clarity is U-shaped: $1 - \exp[-(|z|/1.0)^2]$, so it is lowest near the cutoff (maximally ambiguous) and approaches 1 in the extremes (maximally clear). Coordination is $\Lambda(-0.6\,z) = 1/(1 + e^{0.6z})$, decreasing in z: weak-regime signals ($z < 0$) imply a more open coordination climate. All three jointly determine phrase selection across eight evidence domains.

Figure A12 shows complete briefings generated at $z \in \{-2, 0, +2\}$. At the extremes ($|z| = 2$), clarity is high (0.98) and the cue balance is strongly directional. At $z = 0$, clarity is 0.00 (maximally ambiguous) and the briefing presents a balanced picture.

F Implementation Details

All LLM calls use temperature = 0.7, `max_tokens` = 512, single sample per decision, routed through OpenRouter. Model versions are those available December 2024–February 2025.

Each country has a base prior $\bar{z} \sim \mathcal{N}(0, 0.3)$; each period perturbs it by $\mathcal{N}(0, 0.05^2)$. Regime strength is $\theta \sim \mathcal{N}(\bar{z}, 1)$; private signals are $x_i = \theta + \varepsilon_i$ with $\sigma = 0.3$. The communication network is Watts–Strogatz ($k = 4$, $p = 0.3$), regenerated each period. All random draws use a master seed logged in per-run JSON manifests; LLM responses are cached by request hash for exact reproducibility.

Responses are parsed for JOIN/STAY tokens. Combined error rates are below 2% for five of seven models; Trinity Large has elevated API errors ($\approx 9\%$) due to content filtering. All statistics use `join_fraction_valid` (Table A20).

The dissent floor parameter (default 0.25) controls the minimum probability of contrary-valence cues in any briefing. With the logistic direction slider (slope = 0.8), the valence-channel separation between extreme signals ($z = \pm 2$) is $0.664 \cdot (1 - 2 \cdot \text{floor})$: at the default 0.25, the valence channel retains 50% of the floor-free maximum; halving to 0.125 retains 75%; at 0.50 the valence channel

Table A12: Agent-Level Regressions

	(1) Join Decision Logit		(2) Coord. Ablation Logit		(3) Belief → Action Logit	
θ	-1.741^{***}	(0.043)				
Direction			-14.088^{***}	(0.996)		
Coordination			-9.333^{***}	(1.131)		
Dir × Coord			-0.042	(0.360)		
Belief					0.367^{***}	(0.021)
z-score					-0.040	(0.077)
Comm	0.024	(0.023)				
Flip	0.017	(0.055)				
Propaganda K10	-0.388^{***}	(0.069)				
Propaganda K2	-0.237^{***}	(0.073)				
Propaganda K5	-1.049^{***}	(0.068)				
Propaganda Surveillance	-1.551^{***}	(0.062)				
Scramble	0.088^{**}	(0.038)				
Surveillance	-1.545^{***}	(0.048)				
$\theta\times$ Comm	-0.465^{***}	(0.041)				
$\theta\times$ Flip	3.304^{***}	(0.071)				
$\theta\times$ Propaganda K10	-1.023^{***}	(0.107)				
$\theta\times$ Propaganda K2	-1.115^{***}	(0.113)				
$\theta\times$ Propaganda K5	-1.942^{***}	(0.091)				
$\theta\times$ Propaganda Surveillance	-0.260^{***}	(0.082)				
$\theta\times$ Scramble	1.699^{***}	(0.044)				
$\theta\times$ Surveillance	-1.153^{***}	(0.062)				
Treat: Propaganda K5					0.009	(0.244)
Treat: Pure					4.819^{***}	(1.106)
Treat: Surveillance					0.028	(0.253)
Constant	-0.101^{**}	(0.050)	11.022^{***}	(1.065)	-26.495^{***}	(1.581)
Model FE	Yes		No		No	
Clustered SE	Yes		Yes		Yes	
N	289,555		44,662		18,990	
Pseudo R^2	0.381		0.442		0.975	

Notes: Logit coefficients reported with clustered standard errors (model–country–period) in parentheses. Column (1): agent-level join decision on θ, treatment dummies, and interactions, with model fixed effects. Base category: pure treatment. Column (2): coordination ablation—$\Pr(\text{join}_i = 1) = \Lambda(\beta_0 + \beta_1 \text{Dir}_i + \beta_2 \text{Coord}_i + \beta_3 \text{Dir}_i \times \text{Coord}_i)$—using briefing slider values (pure treatment only). Direction $\in [0, 1]$ is increasing in regime strength; Coordination $\in [0, 1]$ is decreasing (high = weaker regime). Because both are monotone logistic transforms of the same signal, they are near-collinear ($\rho \approx -1$); individual coefficients reflect the partial effect *conditional on the other*, not the marginal effect. Column (3): partial effect of elicited belief ($b_i \in [0, 100]$, stated success probability) on action, controlling for z-score. "Treat: X" rows are treatment intercept dummies (base category: communication), not interactions with belief. **Caveat:** beliefs are elicited *after* the join/stay decision (post-decision), so the belief coefficient reflects association rather than a causal pathway; post-decision rationalization may inflate the correlation. The very high pseudo-R^2 (0.975) and large intercept indicate near-deterministic prediction consistent with quasi-separation—belief almost perfectly predicts action, as expected when belief is stated *post hoc*. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A13: Logistic fit parameters by model and treatment. Fitted form: $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$, where $b_1 > 0$ implies a *decreasing* join curve. $\hat{\theta}^*$ is the estimated cutoff $(-b_0/b_1)$; $\beta \equiv b_1$ is the steepness parameter.

	Pure		Communication	
Model	$\hat{\theta}^*$ (SE)	β (SE)	$\hat{\theta}^*$ (SE)	β (SE)
Mistral Small Creative	−0.32 (0.02)	+2.15 (0.08)	−0.23 (0.02)	+2.57 (0.11)
Llama 3.3 70B	+0.02 (0.04)	+2.83 (0.36)	−0.06 (0.04)	+3.63 (0.52)
Qwen3 30B	+0.10 (0.06)	+1.62 (0.16)	−0.03 (0.04)	+2.94 (0.36)
GPT-OSS 120B	−0.25 (0.04)	+2.06 (0.15)	−0.15 (0.03)	+3.03 (0.26)
Qwen3 235B	−0.22 (0.03)	+2.11 (0.15)	−0.22 (0.03)	+2.62 (0.23)
Trinity Large	+0.08 (0.05)	+1.34 (0.11)	−0.02 (0.04)	+2.23 (0.23)
MiniMax M2-Her	−0.17 (0.09)	+0.61 (0.06)	−0.07 (0.09)	+0.66 (0.06)

Notes: Fitted form: $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$, so $b_1 > 0$ corresponds to a decreasing join curve (the expected direction). $\hat{\theta}^* = -b_0/b_1$: estimated cutoff. $\beta \equiv b_1$: logistic steepness (larger = sharper threshold). Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

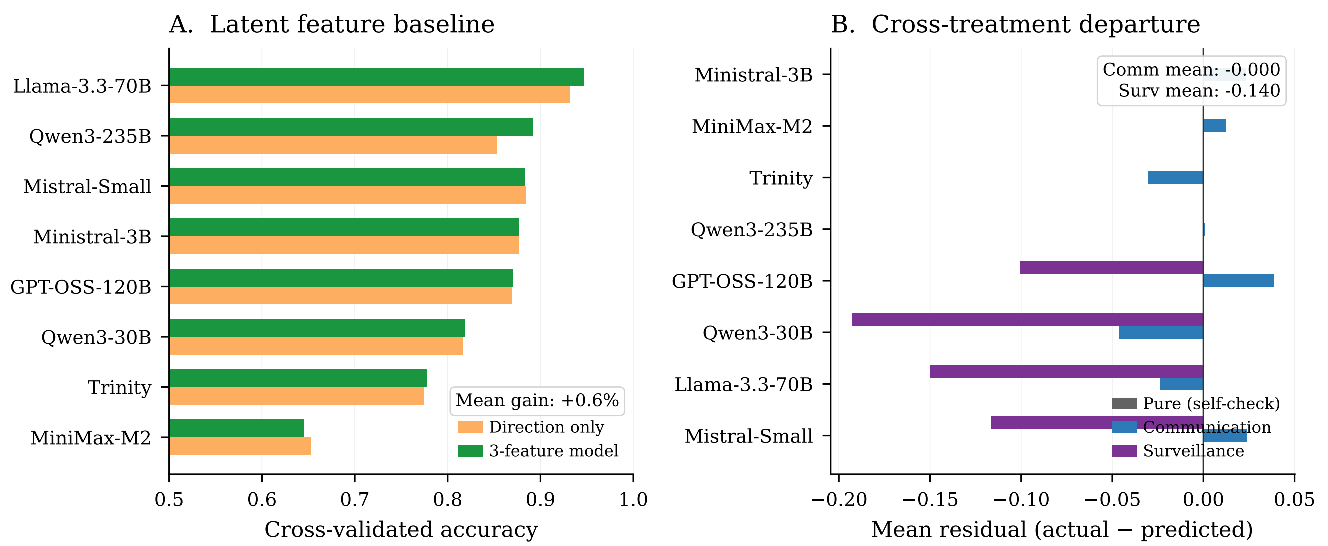


Figure A7: Construct validity tests. (A) Three-feature vs. one-feature prediction accuracy. (B) Cross-treatment generalization.

Table A14: Cross-generator robustness. Three text rendering styles (baseline, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs. The Pearson $r(J, A(\theta))$ and logistic cutoff are virtually identical across generators.

Model	Generator	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small Creative	Baseline	100	0.402	+0.77	-0.070
Mistral Small Creative	Cable	100	0.406	+0.76	-0.057
Mistral Small Creative	Journalistic	100	0.403	+0.75	-0.065
Llama 3.3 70B	Baseline	100	0.354	+0.75	-0.249
Llama 3.3 70B	Cable	100	0.362	+0.75	-0.226
Llama 3.3 70B	Journalistic	100	0.360	+0.76	-0.233

Notes: Information design θ-grid. Three text renderers (baseline intelligence briefing, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs.

Table A15: Agent-level belief correlations (primary model: Mistral Small Creative).

Treatment	N	r_{post}	$r_{\text{b,d}}$	r_{partial}	Mean belief
Pure	10000	+0.78	+0.84	+0.64	0.444
Communication	5000	+0.79	+0.83	+0.56	0.444
Surveillance	5000	+0.78	+0.73	+0.47	0.436

Notes: r_{post}: posterior vs. stated belief. $r_{\text{b,d}}$: belief vs. decision. r_{partial}: belief–decision controlling for z-score.

collapses to zero by construction. Clarity and coordination sliders, which depend on z independently of the floor, continue to carry signal at all floor values—so the threshold relation is not knife-edge in this design choice.

All code, prompts, cached responses, and output data are available at https://github.com/keltokhy/llm-global-games.

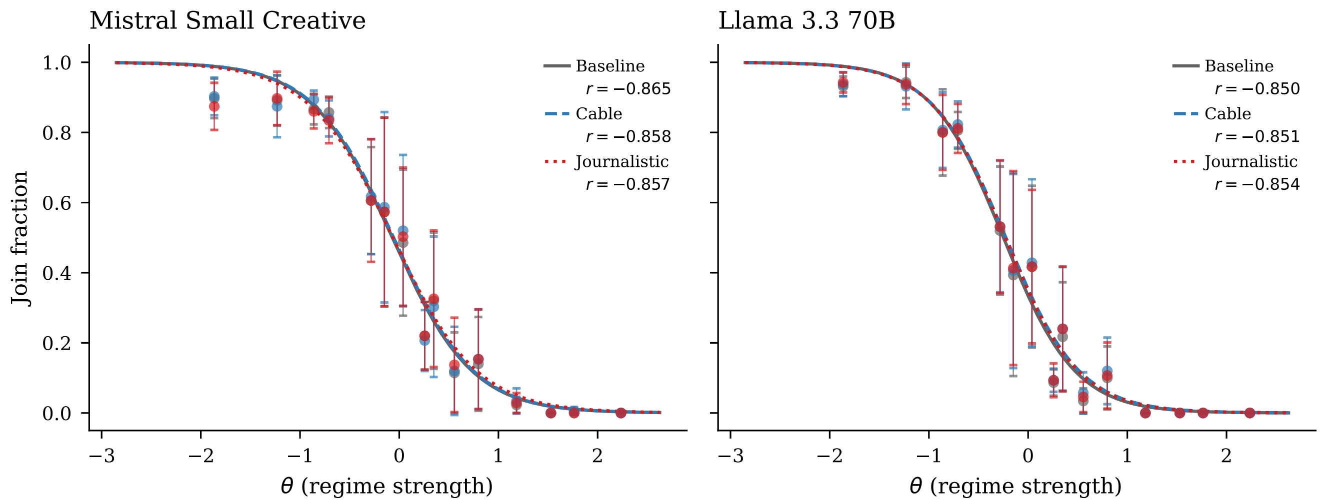


Figure A8: Cross-generator robustness. Three text formats produce indistinguishable sigmoids.

Table A16: Elicited punishment risk (0–10 scale). Agents rate expected regime punishment after their JOIN/STAY decision. "JOIN" and "STAY" columns show the mean rating conditional on the agent's own decision.

Model	Condition	N	Mean risk	Risk | JOIN	Risk | STAY
Mistral	Pure	2500	8.1	8.0	8.2
Mistral	Comm	2500	8.1	8.0	8.1
Mistral	Surveillance	2500	8.1	8.1	8.1
Llama 70B	Pure	2500	8.0	8.0	7.9
Llama 70B	Comm	2500	7.9	8.0	7.9
Llama 70B	Surveillance	2500	7.9	8.0	7.9

Notes: Agent-level elicitation on a 0–10 scale. Reported risk is the mean rating conditional on the agent's own JOIN or STAY decision.

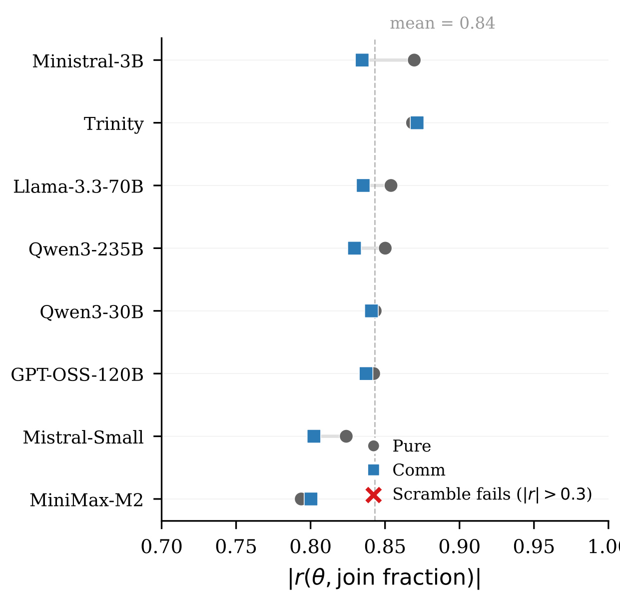


Figure A9: Cross-model signal monotonicity. Points report $r(J, A(\theta))$ under pure and communication.

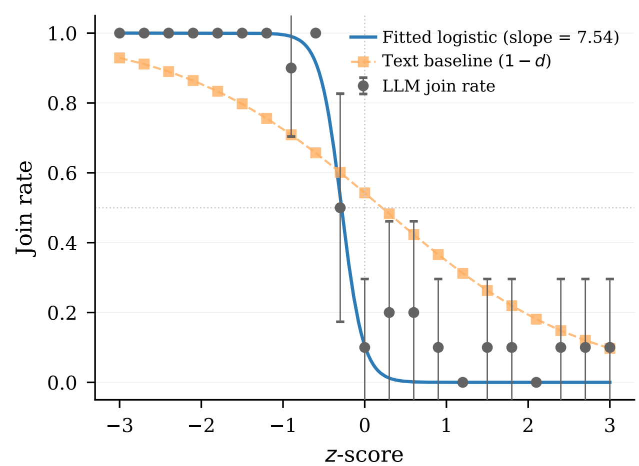


Figure A10: Text baseline test. The LLM join curve is steeper than the naïve text predictor $(1 - d)$, indicating processing beyond surface sentiment.

Table A17: Stakes conditions. *Top:* pipeline-centering B/C sweep—only (B, C) vary. *Bottom:* narrative stakes ladder—only a prepended header varies.

Condition / target θ^*	B	C	Header / narrative framing
Payoff sweep (standard briefings, no header):			
$\theta^* = 0.25$	0.33	1	—
$\theta^* = 0.33$	0.49	1	—
$\theta^* = 0.45$	0.82	1	—
$\theta^* = 0.50$	1.00	1	—
$\theta^* = 0.60$	1.50	1	—
$\theta^* = 0.67$	2.03	1	—
$\theta^* = 0.75$	3.00	1	—
Narrative cost ladder (B=C=1; header prepended):			
Very high cost	1	1	Disappearances, families targeted
Moderate high	1	1	Prison sentences, job loss
Neutral	1	1	Consequences uncertain
Moderate low	1	1	Brief questioning, released
Very low cost	1	1	Guaranteed asylum, zero cost
Controls:			
Placebo	1	1	Weather information (irrelevant)
High cost v2	1	1	Paraphrase of very high cost
Low cost v2	1	1	Paraphrase of very low cost

Notes: Payoff sweep: $B = \theta^*/(1-\theta^*)$, C=1. Z-score center and θ-grid shift with θ^*, so the briefing-text distribution is approximately constant across the seven cells. Narrative ladder: B=C=1 in every row; only the prepended header varies. Join rates: very high cost 21.3%, moderate high 46.6%, neutral 46.7%, moderate low 68.4%, very low 78.9%. Placebo 32.7%, v2 variants 27.4%/63.8%. Full headers in Appendix A.

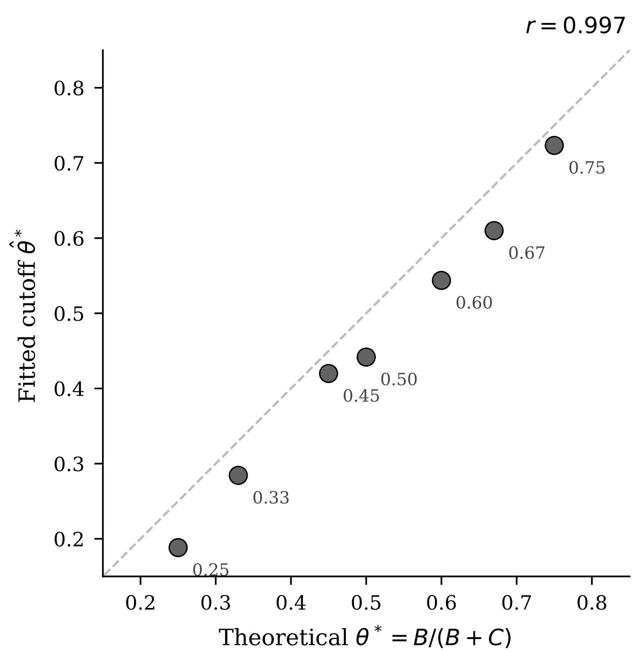


Figure A11: Fitted cutoff $\hat{\theta}^*$ vs. theoretical θ^* across seven B/C ratios ($r = 0.997$). All conditions use standard briefings; the z-score grid shifts with θ^*, so this tests pipeline centering rather than direct payoff sensitivity.

Table A18: Strategic-stakes narrative comparative statics. High-cost framing emphasizes severe reprisals for failed action; low-cost framing emphasizes minimal consequences. The table is evidence on qualitative stakes framing, not literal numeric payoff reasoning.

Design	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$ (SE)	Δ (pp)
Baseline	270	0.409	+0.88	0.39 (0.007)	—
High cost	270	0.190	+0.87	0.13 (0.010)	-21.9
Low cost	270	0.693	+0.88	0.72 (0.007)	+28.4

Notes: Primary model, information design θ-grid. Identical briefings across conditions; only the strategic-stakes header varies. Δ: change vs. baseline mean join (pp).

Table A19: Slider values at representative z-scores.

z	Direction	Clarity	Coordination
−2.0	0.17	0.98	0.77
−1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23

Table A20: Parse error and API failure rates by model and treatment.

Model	Treat.	N	API err	Unparse.	Combined
Mistral	Pure	1000	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	200	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Llama 70B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Qwen 30B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
GPT-OSS	Pure	200	0.0%	1.5%	1.5%
	Comm	200	0.0%	0.3%	0.3%
	Scr.	500	0.0%	3.1%	3.1%
	Flip	500	0.0%	3.3%	3.3%
Qwen 235B	Pure	200	0.0%	0.0%	0.0%
	Comm	200	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Trinity	Pure	100	9.0%	0.0%	9.0%
	Comm	100	10.0%	0.0%	10.0%
	Scr.	100	8.0%	0.0%	8.0%
	Flip	100	9.9%	0.0%	9.9%
MiniMax	Pure	100	0.0%	1.5%	1.5%
	Comm	100	0.0%	0.9%	0.9%
	Scr.	100	0.0%	1.8%	1.8%
	Flip	100	0.0%	1.0%	1.0%

Notes: Per-period averages. API error = provider-side failure (timeout, rate limit, content filter). Unparseable = valid response not classified as JOIN/STAY. Combined < 2% for five of seven models; Trinity Large has elevated API errors (≈9%) due to content filtering.

$z = -2$ **(weak regime)**

PRIVATE BRIEFING

BOTTOM LINE: Signs of institutional erosion are accumulating faster than the regime can manage them.
ATMOSPHERE: Coordination is happening in plain sight — meetings, signals, preparations that would have been unthinkable a month ago.

EVIDENCE:

- - [Elite Cohesion] senior figures are giving interviews that subtly distance themselves from the leader's recent decisions
- - [Security Forces] reports of soldiers fraternizing with civilians more than usual at checkpoints (overwhelming convergence — every channel telling the same story independently)
- - [Money And Logistics] customs and trade are flowing normally at the borders — no unusual delays or inspections (overwhelming convergence — every channel telling the same story independently)
- - [Street Mood] jokes about the regime are being told openly in places where that would have been unthinkable
- - [Information Control] leaked documents are circulating faster than the regime can discredit them (overwhelming convergence — every channel telling the same story independently)
- - [Personal Observations] your cousin in the army hasn't called in two weeks — unusual for someone who calls every Sunday (near-unanimous confirmation from every reliable source)
- - [Diplomatic Signals] international shipping companies have reclassified the country as higher-risk for cargo insurance — the trend cooling off, though not yet cold
- - [Institutional Functioning] municipal services — garbage collection, water, permits — have become erratic in the capital

UNCERTAINTIES:

- - Whether any of the apparent weakness is being deliberately staged to flush out opponents.
- - How much of the disarray is structural versus a temporary response to a specific crisis.

$z = 0$ **(ambiguous)**

PRIVATE BRIEFING

BOTTOM LINE: The situation could tip either way — there are credible signals pointing in both directions.
ATMOSPHERE: There's a palpable sense that a critical mass is forming, even if nobody can count the numbers yet.

EVIDENCE:

- - [Elite Cohesion] elites are publicly unified but privately positioning themselves for multiple outcomes (a couple of contacts mentioning it independently)
- - [Security Forces] the security forces are maintaining presence but seem to be conserving resources rather than demonstrating strength (a couple of contacts mentioning it independently)
- - [Money And Logistics] economic indicators are mixed — retail spending is down but construction permits are up
- - [Street Mood] people are frustrated but fatalistic — complaining more, but not in a way that suggests action
- - [Information Control] the regime is still shaping the narrative but has lost the initiative — they're responding to events rather than framing them
- - [Personal Observations] a well-connected friend said 'it could go either way' and seemed genuinely unsure (a couple of contacts mentioning it independently)
- - [Diplomatic Signals] foreign embassies are maintaining full operations but have updated contingency plans — standard prudence or genuine concern
- - [Institutional Functioning] the system is creaking but not collapsing — people are working around the bottlenecks rather than through them (a couple of contacts mentioning it independently)

UNCERTAINTIES:

- - Whether the population's quiescence reflects genuine acceptance or merely the absence of a triggering event.
- - Whether the stability you're seeing is structural or just a temporary equilibrium that could shift quickly.

$z = +2$ **(strong regime)**

PRIVATE BRIEFING

BOTTOM LINE: Current indicators point to maintained control, with internal challenges being managed rather than spiraling.
ATMOSPHERE: People are aware of each other's frustrations but nobody is naming them directly. It's communication through omission.

EVIDENCE:

- - [Elite Cohesion] the regime successfully managed a minor scandal by presenting a unified response within hours
- - [Security Forces] security checkpoints have been reduced, suggesting confidence rather than siege mentality (overwhelming convergence — every channel telling the same story independently)
- - [Money And Logistics] customs and trade are flowing normally at the borders — no unusual delays or inspections (overwhelming convergence — every channel telling the same story independently)
- - [Street Mood] people complain but do it like people who expect tomorrow to look much like today
- - [Information Control] social media monitoring appears sophisticated and well-resourced (overwhelming convergence — every channel telling the same story independently)
- - [Personal Observations] your cousin in the army hasn't called in two weeks — unusual for someone who calls every Sunday (near-unanimous confirmation from every reliable source)
- - [Diplomatic Signals] foreign students continue to enroll at local universities — their embassies haven't advised them to leave — the trend moving with a momentum that would be hard to reverse
- - [Institutional Functioning] municipal services are running smoothly — a mundane but reliable indicator of institutional health

UNCERTAINTIES:

- - Whether the regime's strength is being accurately perceived by potential challengers — overestimation could be self-fulfilling.
- - If unrest flares, the response capacity seems intact — though no system is invulnerable to surprises.

Figure A12: Example briefings at three z-scores. *Left:* $z = -2$ (strong join signal) — evidence of elite fracturing, security friction, and capital flight. *Center:* $z = 0$ (ambiguous) — balanced mixed signals across all domains. *Right:* $z = +2$ (strong stay signal) — evidence of elite cohesion, military confidence, and economic stability. The displayed text is an unedited generator draw at seed 5150, agent 8, period 124 (aside from line wrapping).